Zikun Wang

857-300-7432 | zikunwang00@gmail.com | www.zikunw.com | github.com/zikunw | linkedin.com/in/zikunw

Research Interests

High-performance data systems: GPU-accelerated vector search and approximate nearest neighbor indexing, I/O-efficient data structures, and reconfigurable stream processing systems.

Education

Northeastern University — Boston, MA
Ph.D. in Computer Science — Advisor: Prof. Prashant Pandey (DATA Lab) — *Sep. 2025 – Present*

Boston University — Boston, MA
B.A. and M.S. in Computer Science — *Jan. 2021 – Dec. 2024*

Conference Publications

1. **Koala: Non-disruptive Reconfiguration for Long-running Stateful Dataflows.** Lei Huang, Sakshi Sharma, Chenxu Yu, **Zikun Wang**, Yuanli Wang, Zichen Zhu, John Liagouris, and Vasiliki Kalavri. *USENIX Symposium on Networked Systems Design and Implementation* (NSDI 2027). To appear.
2. **The Dangerous Impact of Solver Imprecision on Data Management Techniques (and How to Avoid It).** Zixuan Chen, **Zikun Wang**, Panagiotis Manolios, and Mirek Riedewald. *International Conference on Extending Database Technology* (EDBT 2027). To appear.
3. **GPU-Accelerated ANNS: Quantized for Speed, Built for Change.** Hunter McCoy, **Zikun Wang**, and Prashant Pandey. *Proceedings of the VLDB Endowment* (VLDB 2026). arXiv:2601.07048
4. **CAPSys: Contention-aware Task Placement for Data Stream Processing.** Yuanli Wang, Lei Huang, **Zikun Wang**, Vasiliki Kalavri, and Ibrahim Matta. *European Conference on Computer Systems* (EuroSys 2025). doi:10.1145/3689031.3696085

Workshop Papers

1. **Directional Beam Search: I/O-Efficient ANNS on GPU. Zikun Wang**, Hunter McCoy, and Prashant Pandey. *2nd Workshop on Vector Databases* (VecDB 2026).

Software

Jasper | github.com/saltsystemslab/jasper — 2025 – Present

- Lead developer of an open-source GPU library for billion-scale approximate nearest neighbor search, supporting in-place updates and indexes larger than GPU memory.

Research Experience

DATA Lab, Northeastern University — Boston, MA
Research Assistant — Advisor: Prof. Prashant Pandey — *Sep. 2025 – Present*

Complex Analytics & Scalable Processing Lab (CASP), Boston University — Boston, MA
Research Assistant — Advisors: Prof. Vasiliki Kalavri, Prof. John Liagouris — *Sep. 2023 – Aug. 2025*

Research Projects

Updatable and Scalable GPU Approximate Nearest Neighbor Index — May 2025 – Present

Advisor: Prof. Prashant Pandey, Northeastern University

- Designed and implemented *Jasper*, a GPU vector index that supports in-place updates and scales beyond GPU memory, enabling billion-scale approximate nearest neighbor search.
- Achieved faster index construction and query throughput with a smaller memory footprint than state-of-the-art libraries including NVIDIA cuVS.
- Developed *directional beam search*, an I/O-efficient graph traversal strategy that reduces host-to-device transfer volume during search on out-of-memory indexes.

A Disaggregated Stream Processing System for Heterogeneous Environments — Jan. 2024 – Aug. 2025

Advisors: Prof. Vasiliki Kalavri, Prof. John Liagouris, Boston University

- Proposed a fully disaggregated design for stream processing that decouples data-plane and control-plane services, enabling non-blocking online reconfiguration without pausing computation.
- Implemented a distributed streaming runtime with a Flink-like dataflow API, together with a full test suite and a metric collection system for local and cloud deployments; integrated telemetry with Prometheus for real-time observability.

Automatic Task Placement for Apache Flink Oct. 2023 – May 2024

Advisor: Prof. Vasiliki Kalavri, Boston University

- Implemented *CAPSys*, a task placement controller for Apache Flink that delegates dataflow tasks to physical resources for data stream processing.
- Designed an adaptive placement policy that balances compute-, I/O-, and network-intensive tasks across available resources, achieving orders-of-magnitude lower computation time and up to 6× higher throughput than prior work at fixed resources.

Talks and Presentations

1. **Jasper: ANNS Quantized for Speed, Built for Change on GPU** North East Database Day (NEDB Day) 2026. Jan. 2026.
2. **Zero-downtime Reconfiguration Mechanisms for Dynamic Data Stream Processing.** North East Database Day (NEDB Day) 2025. Jan. 2025.
3. **Effective Workload Scaling in Streaming Systems.** 27th Annual Undergraduate Research Symposium, Boston University. Oct. 2024.
4. **Automatic Task Placement Decisions for High-throughput and Resource-efficient Data Stream Processing.** North East Database Day (NEDB Day) 2024. May 2024.
5. **A Computational Platform for Visuospatial Assessment.** Neuroscience of the Everyday World Conference 2024. Aug. 2024.

Teaching

Teaching Assistant | GRS CS 630 Graduate Algorithms | Boston University | Fall 2024
Course Grader | CAS CS 320 Concepts of Programming Languages | Boston University | Fall 2022

Technical Skills

Languages: CUDA, C++, Rust, Go, Java, Python, TypeScript/JavaScript, R.
Systems & Tools: Apache Flink, Kafka, Storm, PostgreSQL, gRPC, Docker, Prometheus, NVIDIA cuVS, React.